Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Unphysicality of Hilbert spaces for quantum mechanics

Files.

- File: `briefs/HilbertUnphysicality/HilbertUnphysicality.lean`
- Library: `HilbertUnphysicality`
- Imports: `InconsistentConvergence`, one new `ForPhyslib` file:
  `ForPhyslib/QuantumMechanics/UnitaryEquivalence.lean` (a unitary carries the quantum model),
  and ten new `ForMathlib` files:
  `ForMathlib/Analysis/InnerProductSpace/l2Diagonal.lean` (diagonal operators on ℓ^2),
  `ForMathlib/Analysis/InnerProductSpace/l2DiagonalEvolution.lean` (their evolution),
  `ForMathlib/Analysis/SpecialFunctions/Gaussian/Erf.lean` (the error function),
  `ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (separable Hilbert spaces are $\ell^2(\mathbb{N})$),
  `ForMathlib/MeasureTheory/Function/L2Space.lean` (L^2 is infinite-dimensional),
  `ForMathlib/MeasureTheory/Function/L2Multiplication.lean` (multiplication operators on L^2 and their evolution),

`ForMathlib/MeasureTheory/Function/LpSpace/ChangeOfVariables.lean` (the unitary of a change of variables on $L^2(\mathbb{R})$),
`ForMathlib/LinearAlgebra/LinearPMap.lean` (composition on the natural domain),
`ForMathlib/Analysis/Normed/Algebra/Wielandt.lean` (the Wielandt–Wintner theorem),
`ForMathlib/Analysis/Distribution/TemperedDistribution.lean` (the Leibniz rule in $\mathcal{S}'$)

Setting. Theorem 5 says that the Hilbert-space formulation QSTO:H is inconsistent with three physical conditions. The brief proves four mathematical propositions (Sec. 2) and then argues each part of Theorem 5 from one of them (Sec. 4). The propositions are checked in Lean, and Theorem 5 is stated and proved as `theorem5`, with each condition in its minimum form (see the open problems, "Precise readings"). Of the additional remarks of Sec. 5, the one on the domain of the Schrödinger equation is checked too, for the harmonic oscillator in the number basis and for the free particle in position space, and so is the one on commutator relations, for position and momentum on $L^2(\mathbb{R})$ (see "Differences").

Results.

Brief	Lean
Proposition 1	
$\mathcal{D}(A) \neq \mathcal{H}$ (Hellinger–Toeplitz)	`bounded_of_isSymmetric`, `not_isSymmetric_of_unbounded`, `domain_ne_top_of_unbounded`
A is not continuous on $\mathcal{D}(A)$	`not_continuous_of_unbounded`
Proposition 2	
N with domain $\mathcal{D}(N)$ is self-adjoint	`numberOp`, `coe_domain_numberOp`, `isSelfAdjoint_numberOp` (from `lp.isSelfAdjoint_diagonal`)
N is unbounded, so not continuous on $\mathcal{D}(N)$	`exists_lt_norm_numberOp`, `not_continuous_numberOp`
$\mathcal{D}(N)$, $E_N(x) = \langle x\|Nx\rangle / \|x\|^2$	`numberDomain`, `expN`, `expN_eq_inner`
$x_j \to e_0$ in $\mathcal{D}(N) \setminus \{0\}$, $E_N(x_j) = 1$, $E_N(e_0) = 0$	`expN_not_continuousWithinAt`, `not_continuousOn_expN` (sequence from the inconsistent-convergence brief)
Proposition 3	
erf, $\|\operatorname{erf}\| < 1$, erf $\to \pm 1$	`Real.erf`, `Real.abs_erf_lt_one`, `Real.tendsto_erf_atTop`
$f(x) = \tan(\frac{\pi}{2} \operatorname{erf} x)$ is increasing	`frameChange`, `strictMono_frameChange`
f is a bijection of $\mathbb{R}$, $g = f^{-1}$, $g' > 0$	`surjective_frameChange`, `frameIso`, `hasDerivAt_frameIso_symm`
$U\psi = \phi$, at the level of densities	`cauchyDensity_frameChange_mul_deriv`: $\|\phi(f(x))\|^2 f'(x) = \|\psi(x)\|^2$
the same, integrated	`arctan_frameChange`: $\arctan(f(x))/\pi = \operatorname{erf}(x)/2$
ψ normalized, $\psi \in \mathcal{D}(X)$	`integral_gaussDensity`, `integrable_sq_mul_gaussDensity`
ϕ normalized, $\phi \notin \mathcal{D}(Y)$	`integral_cauchyDensity`, `not_integrable_sq_mul_cauchyDensity`
the unitary U, $(U\psi)(y) = \psi(g(y))\sqrt{g'(y)}$	`frameUnitary` (from `Lp.compDiffeo`), `coeFn_frameUnitary`
$\mathcal{D}(X) = \mathcal{D}(Y) = \{\psi : x\psi \in L^2\}$	`positionDomain`
$\psi \in \mathcal{D}(X)$, $U\psi = \phi$, $U\psi \notin \mathcal{D}(Y)$	`gaussState_mem_positionDomain`, `frameUnitary_gaussState`, `frameUnitary_gaussState_not_mem`
the minimum form of [B]WDQ fails	`not_forall_frameUnitary_mem_positionDomain`

Brief	Lean
Proposition 4	
separable infinite-dimensional Hilbert spaces are isomorphic	`nonempty_linearIsometryEquiv_of_separable`
$L^2(\mathbb{R}^n)$ is separable and infinite-dimensional	`L2.separableSpace_euclidean`, `L2.not_finiteDimensional_euclidean`
$L^2(\mathbb{R}^{n+1}) \cong L^2(\mathbb{R}^{m+1})$	`L2_euclidean_iso`
Theorem 5, [B]STDIST	
a unitary induces bijections between rays, density operators, self-adjoint operators	`mapRays`, `isDensityOperator_conjStarAlgEquiv_iff`, `bijective_conjLinearPMap`, `isSelfAdjoint_conjLinearPMap_iff`
and preserves the Born rule and the entropy	`born_map`, `IsDensityOperator.vonNeumannEntropy_conjStarAlgEquiv`
$L^2(\mathbb{R}^{n+1})$ and $L^2(\mathbb{R}^{m+1})$ carry the same QSTO:H model	`carriesQSTO`, `exists_carriesQSTO`
Theorem 5	
QSTO:H is inconsistent with [B]COBS, [B]WDQ and [B]STDIST	`theorem5`
Sec. 5, the domain of the Schrödinger equation	
$\mathcal{D}(H) \neq \mathcal{H}$ for the oscillator	`domain_numberOp_ne_top`, `harmonicState_not_mem_numberDomain` ($x_n = 1/n$)
the evolution e^{-itH}, defined on all of $\mathcal{H}$	`numberEvolution` (from `lp.diagonalEvolution`)
on $\mathcal{D}(H)$ the Schrödinger equation holds	`hasDerivAt_numberEvolution` (from `lp.hasDerivAt_diagonalEvolution`)
off $\mathcal{D}(H)$ it holds at no time	`not_differentiableAt_numberEvolution`, `differentiableAt_numberEvolution_iff`
$V(x)\psi(x)$ need not be square-integrable	`not_memLp_sq_mul_frameUnitary_gaussState`, `frameUnitary_gaussState_not_mem`
Sec. 5, in position space	
$P = -i\,d/dx$, $\mathcal{D}(P) = \{\psi : \xi\hat{\psi} \in L^2\}$	`fourierL2`, `momentumOp`, `mem_domain_momentumOp`
$P\psi$ need not be defined	`roughState_not_mem_domain_momentumOp`, `domain_momentumOp_ne_top`
$\mathcal{D}(P) = H^1(\mathbb{R})$, so $\mathcal{F}^{-1}\phi \notin H^1$	`mem_domain_momentumOp_iff_memSobolev`, `roughState_not_memSobolev`
$H_0 = P^2/2$ and e^{-itH_0}, defined on all of $L^2(\mathbb{R})$	`freeHamiltonian`, `freeEvolution` (from `L2.mulOp`, `L2.mulEvolution`)
on $\mathcal{D}(H_0)$ the Schrödinger equation holds	`hasDerivAt_freeEvolution` (from `L2.hasDerivAt_mulEvolution`)
off $\mathcal{D}(H_0)$ it holds at no time	`not_differentiableAt_freeEvolution`, `differentiableAt_freeEvolution_iff`

Brief	Lean
Sec. 5, commutator relations	
no bounded X, P with $[X,P]=i\hbar$	`not_exists_ccr_bounded` (from `commutator_ne_algebraMap`)
nor everywhere-defined symmetric ones	`not_exists_ccr_symmetric`
X, and $[X,P]$ on its natural domain	`positionOp`, `commutatorXP` (from `LinearPMap.compMax`)
$[X,P]$ is not defined on all of $L^2(\mathbb{R})$	`domain_commutatorXP_ne_top`
in $\mathcal{S}'$, X is multiplication by x and $P=-i\,d/dx$	`positionOp_toTemperedDistribution`, `momentumOp_toTemperedDistribution`
$[X,P]\psi=i\psi$ on $\mathcal{D}([X,P])$	`commutatorXP_apply` (from `TemperedDistribution.lineDerivOp_smulLeftCLM`)
$\mathcal{D}([X,P])$ contains the Schwartz functions, so is dense	`toLp_mem_domain_commutatorXP`, `dense_domain_commutatorXP`
all of the above	`commutator_relations`

Differences from the brief.

- *Proposition 1.* Both parts are stated for linear maps, without Mathlib's partially defined operators. Discontinuity is proved for an unbounded linear map on any normed space, which includes $\mathcal{D}(A)$ with the norm inherited from $\mathcal{H}$. Hellinger–Toeplitz is proved for an everywhere-defined symmetric map; self-adjointness is not needed.

- *Proposition 2.* The number operator `numberOp` is the diagonal operator with entries n on $\ell^2(\mathbb{N})$, on its maximal domain, as a partially defined operator (`LinearPMap`); its domain is `numberDomain` $=\{x : \sum_n n^2|x_n|^2<\infty\}$. Self-adjointness is proved for every diagonal operator with real entries (`lp.diagonal`, in `ForMathlib`). The expectation `expN` $x=\sum_n n\,|x_n|^2/\|x\|^2$ is defined on all of ℓ^2 and equals $\mathrm{Re}\,\langle x|Nx\rangle\,/\|x\|^2$ on $\mathcal{D}(N)$ (`expN_eq_inner`). Discontinuity is proved within $\mathcal{D}(N)\setminus\{0\}$ with the subspace topology, as in the minimum form of [B]COBS. The sequence x_j is the brief's.

- *Proposition 3.* The wave functions are $\psi=(e^{-x^2}/\sqrt{\pi})^{1/2}$ (`gaussState`) and $\phi=(1/(\pi(1+y^2)))^{1/2}$, as elements of $L^2(\mathbb{R})$. Both frames use the same space $L^2(\mathbb{R})$ with Lebesgue measure. The position operators are both multiplication by the coordinate, `positionOp`, on their common maximal domain `positionDomain` $=\{\psi : x\psi\in L^2\}$; their self-adjointness is not formalized. That f is strictly increasing and onto $\mathbb{R}$, and that $g=f^{-1}$ has $g'>0$, are proved; smoothness is not, only differentiability, which is all the unitary needs. The unitary U is `Lp.compDiffeo`, proved unitary for any bijection g of $\mathbb{R}$ that is differentiable with a differentiable inverse, with weight $\sqrt{|g'|}$; its inverse V is the same construction for g^{-1}. The operator UXU^{-1} and its domain are not formalized. Mathlib has no error function; `Real.erf` is defined in `ForMathlib`.

- *Proposition 4.* The classification of separable infinite-dimensional Hilbert spaces is new in `ForMathlib`. The positive integers n,m appear as $n+1, m+1$.

- *Theorem 5.* Stated as `theorem5`, a conjunction of three parts, one per condition, each for one particular system: the oscillator (Proposition 2), one spatial degree of freedom (Proposition 3), and any two numbers $n+1$, $m+1$ of continuous degrees of freedom (Proposition 4). "Inconsistent" is read as the failure of the minimum form of the condition for that system.

The [B]COBS part states that N is self-adjoint on $\mathcal{D}(N)$, that $E_N(x) = \operatorname{Re}\langle x|Nx\rangle / \|x\|^2$ there, and that E_N is not continuous on $\mathcal{D}(N) \setminus \{0\}$. The [B]WDQ part states $\psi \in \mathcal{D}(X)$, $U\psi = \phi$ a.e. and $U\psi \notin \mathcal{D}(Y)$, for the specific unitary U of the change of frame (for an arbitrary unitary the statement would be trivial). The [B]STDIST part states that a unitary $L^2(\mathbb{R}^{n+1}) \to L^2(\mathbb{R}^{m+1})$ exists and carries the QSTO:H model. For [B]STDIST, that a unitary carries rays, density operators, self-adjoint operators, the Born rule and the entropy is proved for any unitary between complex Hilbert spaces (`carriesQSTO`). Rays are elements of `Projectivization`; density operators and the entropy are those of `ForPhyslib` (positive, trace class, trace one; entropy in $[0, \infty]$); self-adjoint operators are partially defined (`LinearPMap`). The closing remarks on classical cutoffs and on Schwartz spaces are not formalized.

- *Sec. 5, the domain of the Schrödinger equation.* The brief says that the equation is valid only on $\mathcal{D}(H)$. This is read dynamically: the evolution e^{-itH} is defined on all of $\mathcal{H}$, but its orbit $t \mapsto e^{-itH}\psi$ satisfies $i\,\frac{d}{dt}\psi(t) = H\psi(t)$ exactly when $\psi \in \mathcal{D}(H)$, and is differentiable at no time otherwise. It is proved for the harmonic oscillator in the number basis, $H = N$ on $\ell^2(\mathbb{N})$, in units with $\hbar\omega = 1$ and without the zero-point energy, which only adds a global phase. The general statement is proved for every real diagonal operator on ℓ^2 (`lp.diagonalEvolution`, in `ForMathlib`), and for every real multiplication operator on $L^2(\mu)$ (`L2.mulEvolution`). In position space, the momentum operator is $P = \mathcal{F}^{-1}(2\pi\xi)\mathcal{F}$ with Mathlib's unitary Fourier transform on $L^2(\mathbb{R})$, whose convention $\hat{\psi}(\xi) = \int e^{-2\pi i x\xi}\psi(x)\,dx$ gives the factor 2π ($\hbar = 1$). Its maximal domain is identified with $H^1(\mathbb{R})$ in Mathlib's definition (`MemSobolev 1 2`, through the Bessel potential); that this is the space of L^2 functions whose distributional derivative is in L^2 is not proved here. The state outside $\mathcal{D}(P)$ is $\mathcal{F}^{-1}\phi$, for the Cauchy packet ϕ of Proposition 3. The Schrödinger equation is checked in position space for the free particle, $H_0 = \mathcal{F}^{-1}(2\pi^2\xi^2)\mathcal{F}$ (mass 1); that $H_0 = P^2/2$ as partially defined operators is not proved. For a potential $V \neq 0$, see the open problems. For the potential part, $V(y) = y^2$ and the state $\phi = U\psi$ of Proposition 3 are used: $\phi \in L^2$, but $V\phi \notin L^2$; for $V(y) = y$ this is `frameUnitary_gaussState_not_mem`.

- *Sec. 5, commutator relations.* The brief says that $[X, P]/(i\hbar) = I$ cannot hold on the whole of $\mathcal{H}$, since the left side is unbounded, and must be restricted to a domain. Both halves are proved, with $\hbar = 1$. The commutator `commutatorXP` is $XP - PX$ on its natural domain $\{\psi \in \mathcal{D}(X) \cap \mathcal{D}(P) : P\psi \in \mathcal{D}(X),\ X\psi \in \mathcal{D}(P)\}$. Mathlib composes partially defined operators only when the range of one lies in the domain of the other, so the composition on the natural domain is new in `ForMathlib` (`LinearPMap.compMax`). That the relation cannot hold everywhere is proved in two forms: $\mathcal{D}([X, P]) \neq L^2(\mathbb{R})$; and, on any nonzero Hilbert space, no everywhere-defined symmetric X, P satisfy $XP - PX = i\hbar I$. The second form is the Wielandt–Wintner theorem (in a nontrivial normed algebra no commutator is a nonzero scalar, `commutator_ne_algebraMap`, in `ForMathlib`) combined with Hellinger–Toeplitz. That the restricted relation holds is proved on all of $\mathcal{D}([X, P])$: in the tempered distributions, X is multiplication by x and $P = -i\,d/dx$, the Leibniz rule $\frac{d}{dx}(xu) = u + x\frac{du}{dx}$ holds (`TemperedDistribution.lineDerivOp_smulLeftCLM`, in `ForMathlib`), and the embedding $L^2 \to \mathcal{S}'$ is injective. Schwartz functions lie in $\mathcal{D}([X, P])$, so it is dense. Closures and self-adjoint extensions are not considered.

- The other additional remark of Section 5, on point-wise values, is not formalized.

Problems found in the brief.

- Cross-references lack their label: the proof of Theorem 5 has "This satisfies the conditions for 2", "As shown in 3" and "By 4" for Propositions 2, 3 and 4.
- "Schwartz spaces present the same problem" is asserted without proof, and without saying which structure on $\mathcal{S}(\mathbb{R}^n)$ is meant (see the open problems).
- QSTO:H still allows any separable Hilbert space, finite-dimensional ones included. This no longer affects Theorem 5, whose proof reads the inconsistency as the failure of a condition for one particular system, but the brief could say so.